

An **algorithm** is a **step-by-step procedure or set of instructions** used to solve a problem or complete a task.

Think of it like a recipe. Just as a recipe tells you how to cook a dish, an algorithm tells a computer (or a person) exactly what steps to follow to solve a problem.

Example 1: Making Tea (Algorithm)

1. Boil water.
2. Add tea leaves or a tea bag.
3. Add sugar (optional).
4. Add milk (optional).
5. Boil for 2–3 minutes.
6. Strain and serve.

This is an algorithm because the steps are performed in a specific order.

Example 2: Finding the Largest Number

Suppose the numbers are: **10, 25, 15**

Algorithm:

1. Assume the first number is the largest.
2. Compare it with the second number.
3. If the second number is larger, update the largest number.
4. Compare the largest number with the third number.
5. Print the largest number.

Output:

25

Characteristics of an Algorithm

- **Input:** Takes zero or more inputs.
- **Output:** Produces at least one result.
- **Definite:** Every step is clear and unambiguous.
- **Finite:** Ends after a limited number of steps.
- **Effective:** Each step is simple enough to be performed.

Example: Even or Odd Algorithm

1. Start.
2. Read a number.

3. Divide the number by 2.
4. If the remainder is 0, print **"Even"**.
5. Otherwise, print **"Odd"**.
6. Stop.

Why are algorithms important?

Algorithms help us:

- Solve problems efficiently.
- Write computer programs.
- Save time and resources.
- Create software, websites, games, and mobile apps.